



Department of ECE

26EC1202E
DDCA

Exp - 3

HALF ADDER, FULL ADDER, HALF
SUBTRACTOR AND FULL SUBTRACTOR

Half adder

- A half adder adds two single-bit binary numbers (A and B) and produces two outputs: Sum and Carry.
- It's called "half" because it doesn't account for any carry coming in from a previous addition.

Truth table

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Boolean expressions

- $\text{Sum} = A \oplus B$ (XOR)
- $\text{Carry} = A \cdot B$ (AND)

Full adder

- A full adder extends the half adder by adding a third input - **Cin** (carry-in) - so it can be chained together to add multi-bit binary numbers.

Truth table

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Boolean expressions

- $\text{Sum} = A \oplus B \oplus \text{Cin}$
- $\text{Cout} = (A \cdot B) + (\text{Cin} \cdot (A \oplus B))$

Half Subtractor

- A half subtractor subtracts one single bit (B) from another (A) and produces two outputs: Difference and Borrow.

Truth table

A	B	Difference	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Boolean expressions

- Difference = $A \oplus B$ (XOR)
- Borrow = $A' \cdot B$ (NOT A, AND B)

Full Subtractor

- A full subtractor extends the half subtractor by adding a third input — Bin (borrow-in) — so it can be chained together for multi-bit binary subtraction.

A	B	Bin	Difference	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Boolean expressions

- Difference = $A \oplus B \oplus \text{Bin}$
- Bout = $A' \cdot B + (A \oplus B)' \cdot \text{Bin}$

Exp 3: Half adder, full adder, half subtractor & full subtractor design

Aim

To design a half adder, full adder, half subtractor & full subtractor using logic gates.

Exp 4: Half adder and full adder

Apparatus Required:

Hardware:

S.No	Name of the apparatus	Quantity
1	Digital IC trainer kit	1
2	XOR Gate IC (7486)	1
6.	Connecting wires	As per requirement

Software:

Logisim 2.7.1

Theory:

Write 1 or 2 paragraphs about the experiment

Exp 4: Gray to Binary and Binary to Gray Converter

Hardware Procedure:

- Connections are made as per the circuit diagram.
- Verified the connections carefully and switched on the power supply.
- By applying the inputs, the LED outputs are observed and the operation is verified with the help of truth table.

Software Procedure:

- Go to the gates folder and select required logic gates in Logisim.
- Connections are made as per the circuit diagram.
- Assign the input and output nodes.
- Go to simulate option by varying input transitions 0 to 1 and vice versa.
- Check out the Output Logic.

Exp 3: Half adder

Logic diagram:

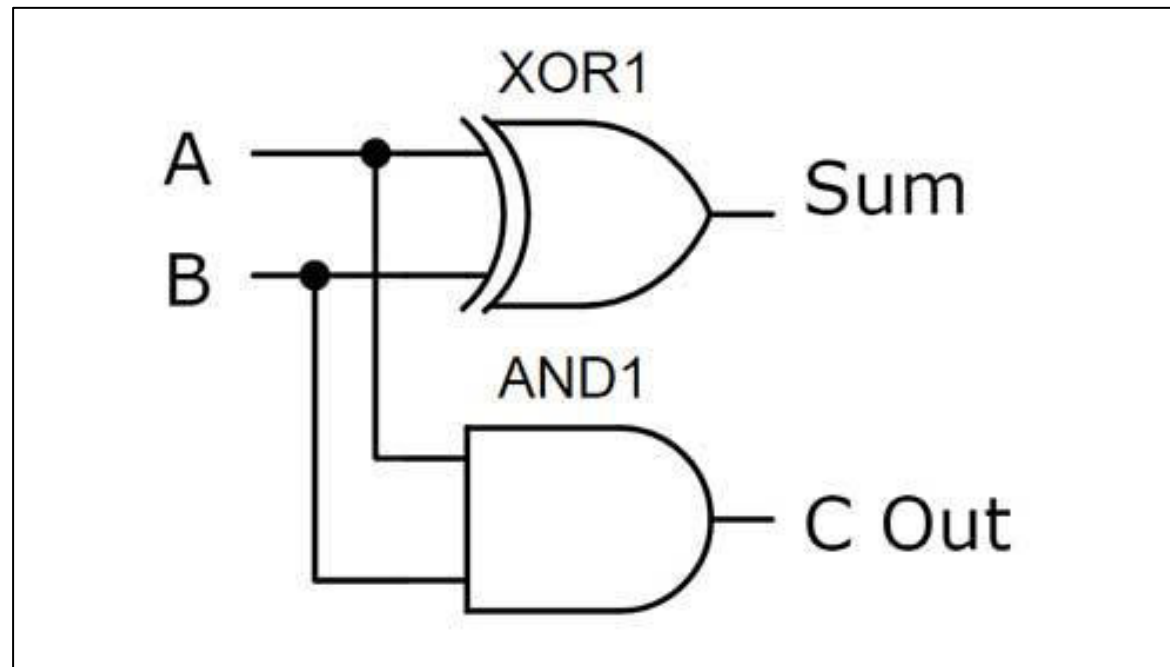


Figure: Logic diagram of half adder

Exp 3: Full adder

Logic diagram:

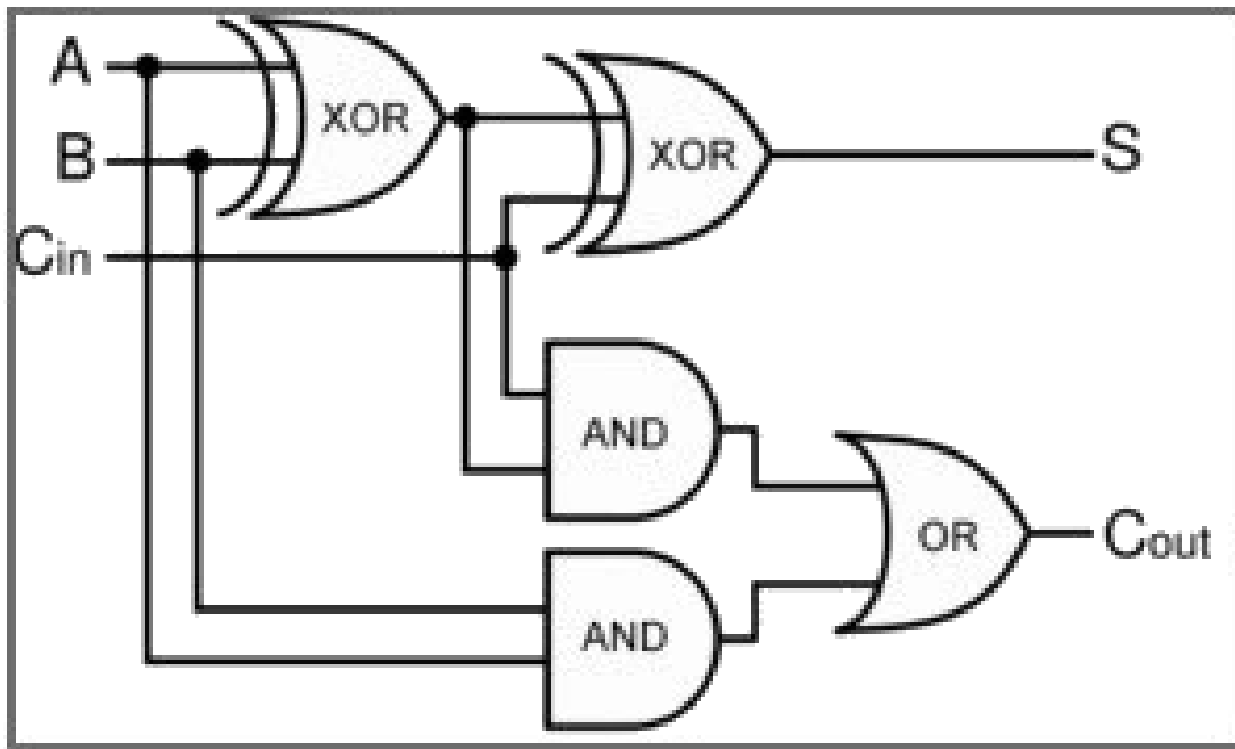


Figure: Logic diagram of full adder

Exp 3: Half subtractor

Logic diagram:

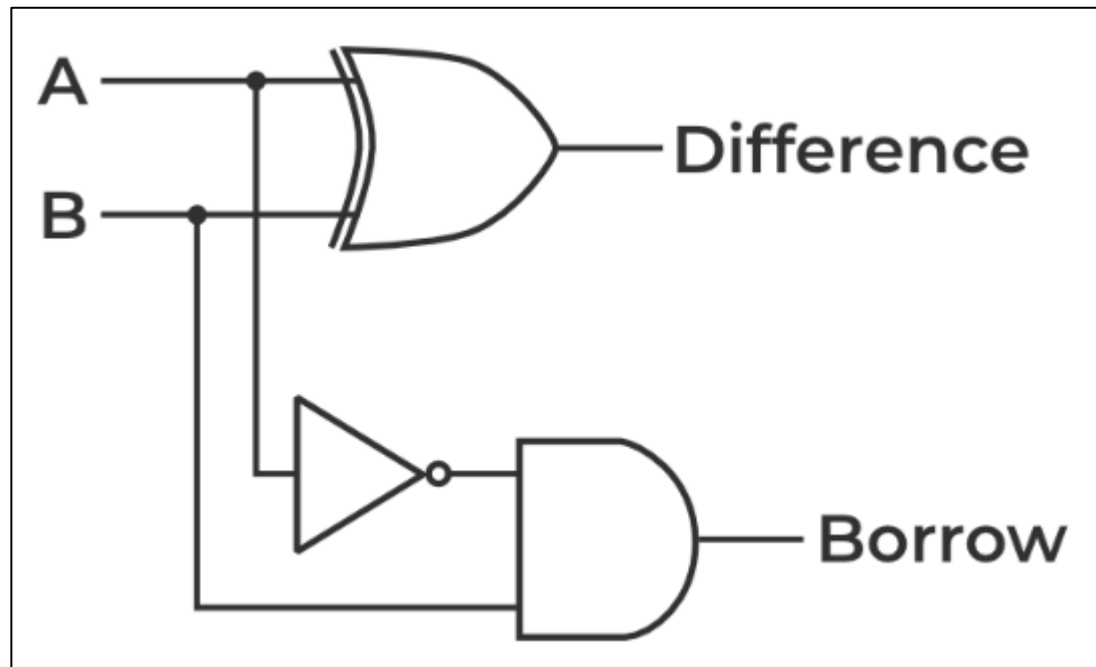


Figure: Logic diagram of Half subtractor

Exp 3: Full subtractor

Logic diagram:

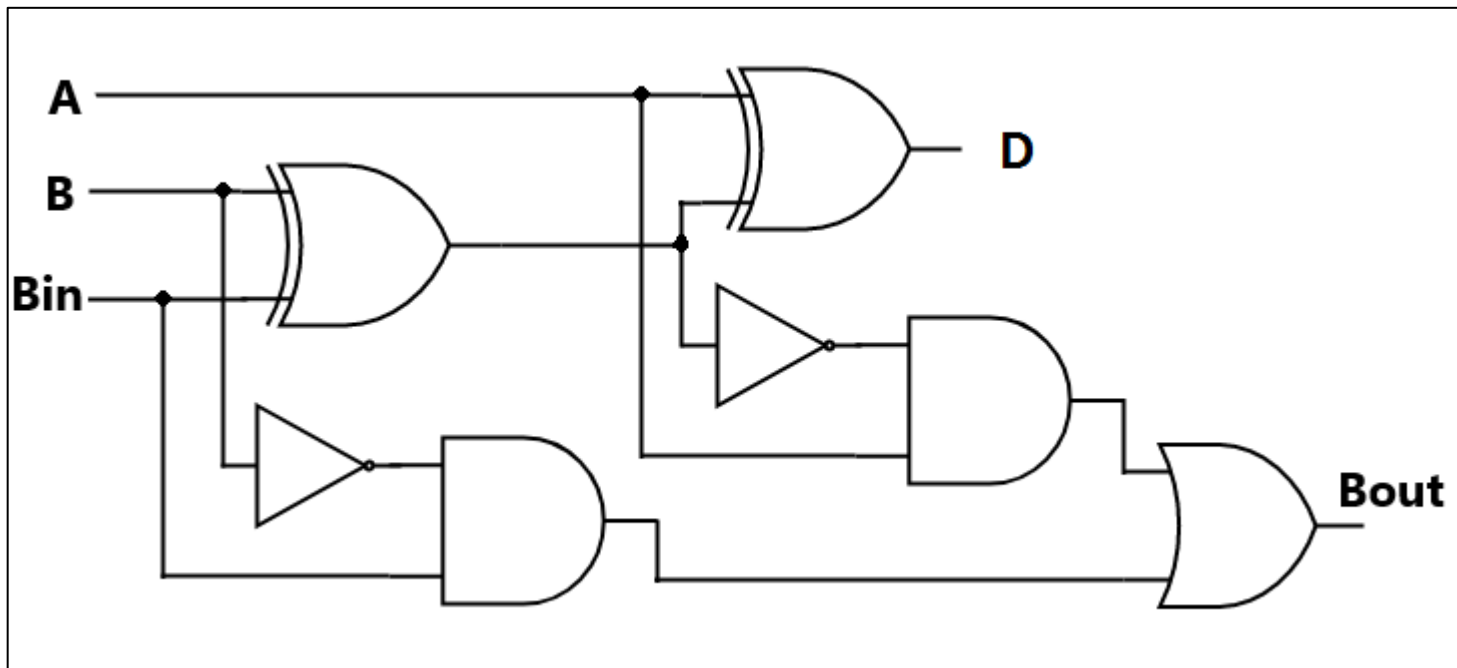


Figure: Logic diagram of full subtractor

THANK YOU



Team – Digital System Design